

Implement SSL-1 (Task-4)

Group-9

Group Members

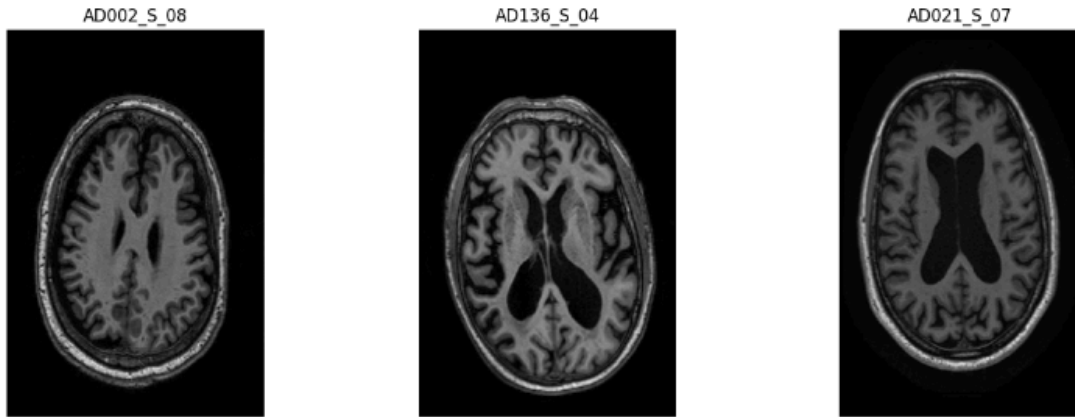
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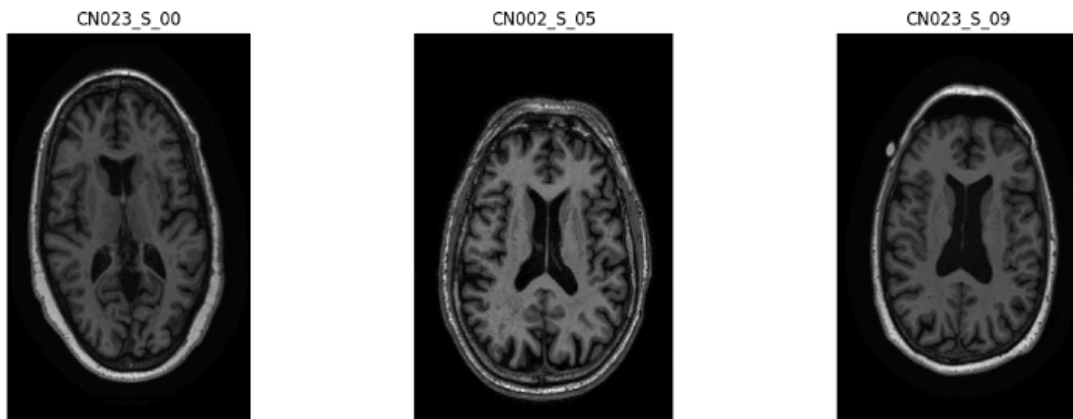
Test Dataset Samples (AD, CN, MCI)

AD (Test)



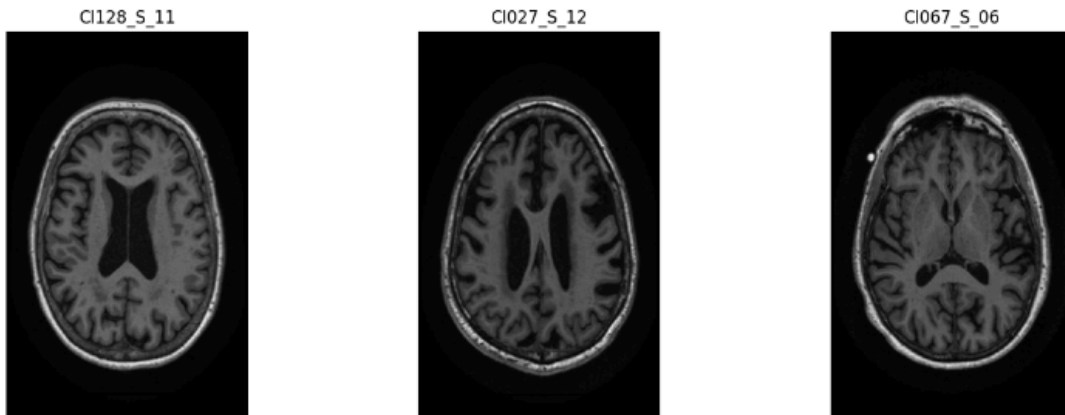
This image represents a brain scan from the test dataset labeled as **Alzheimer's Disease (AD)**. It typically shows structural abnormalities such as cortical thinning and enlarged ventricles. Used to evaluate model performance in identifying AD cases.

CN (Test)



A test image labeled as **Cognitively Normal (CN)**. Displays a healthy brain structure with no visible signs of degeneration. Serves as a baseline for comparison with AD and MCI scans.

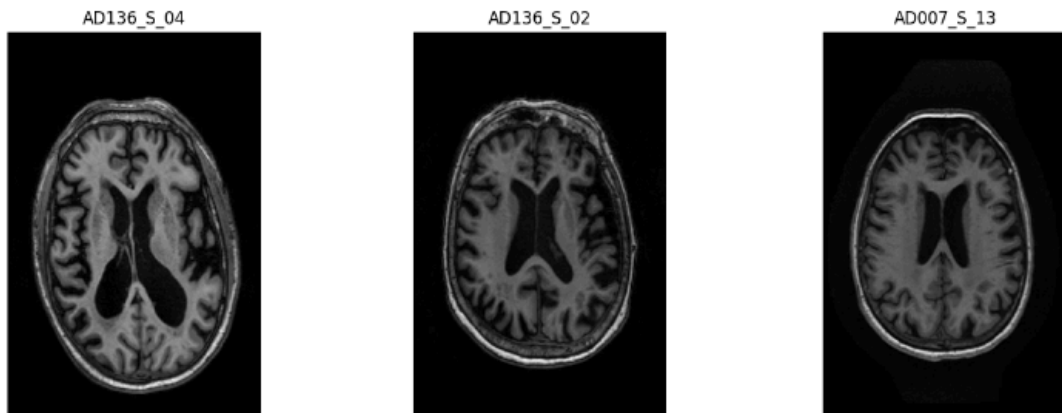
MCI (Test)



This test image is labeled as **Mild Cognitive Impairment (MCI)**. Shows subtle changes in brain structure, intermediate between CN and AD. Helps assess the model's sensitivity to early-stage cognitive decline.

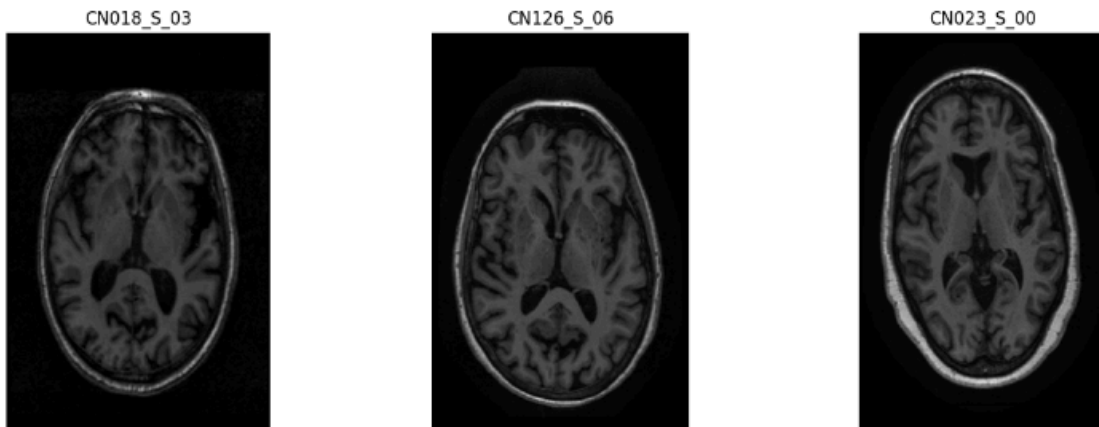
Train Dataset Samples (AD, CN, MCI)

AD (Train)



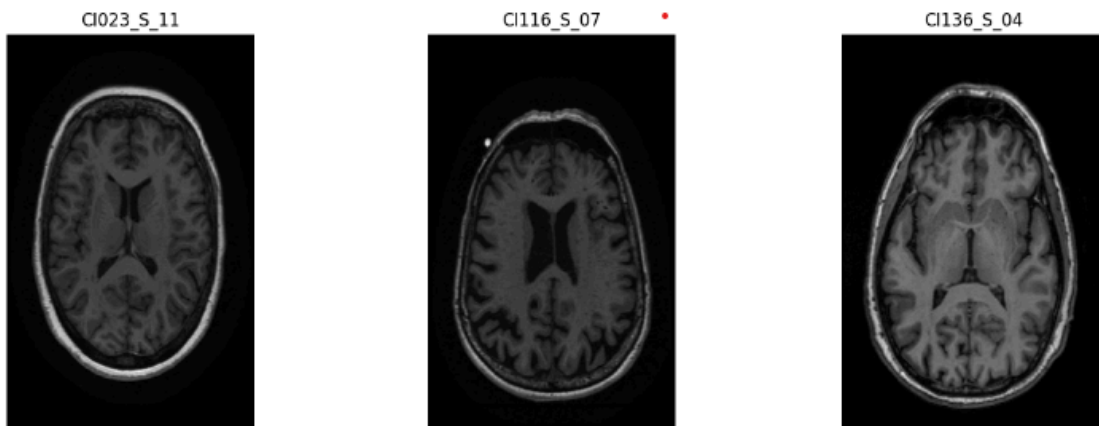
A training image labeled as **Alzheimer's Disease (AD)**. Used to teach the model to recognize patterns associated with AD. Features similar to the AD test image, aiding in model generalization.

CN (Train)



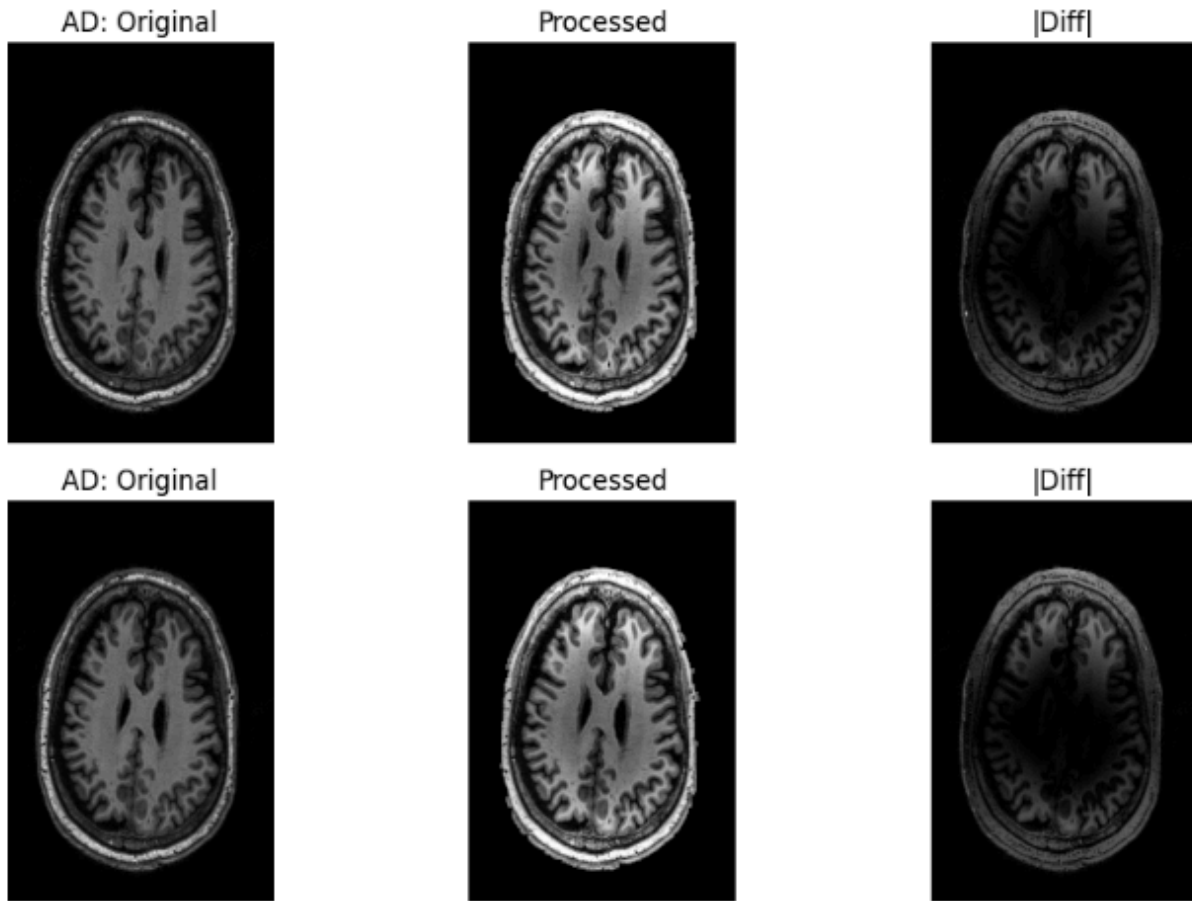
A training image labeled as **Cognitively Normal (CN)**. Represents normal brain anatomy, helping the model learn what healthy scans look like. Crucial for distinguishing CN from MCI and AD.

MCI (Train)



A training image labeled as **Mild Cognitive Impairment (MCI)**. Provides examples of early cognitive decline for model learning. Important for improving classification accuracy between CN and AD.

AD Original- Processed- Difference

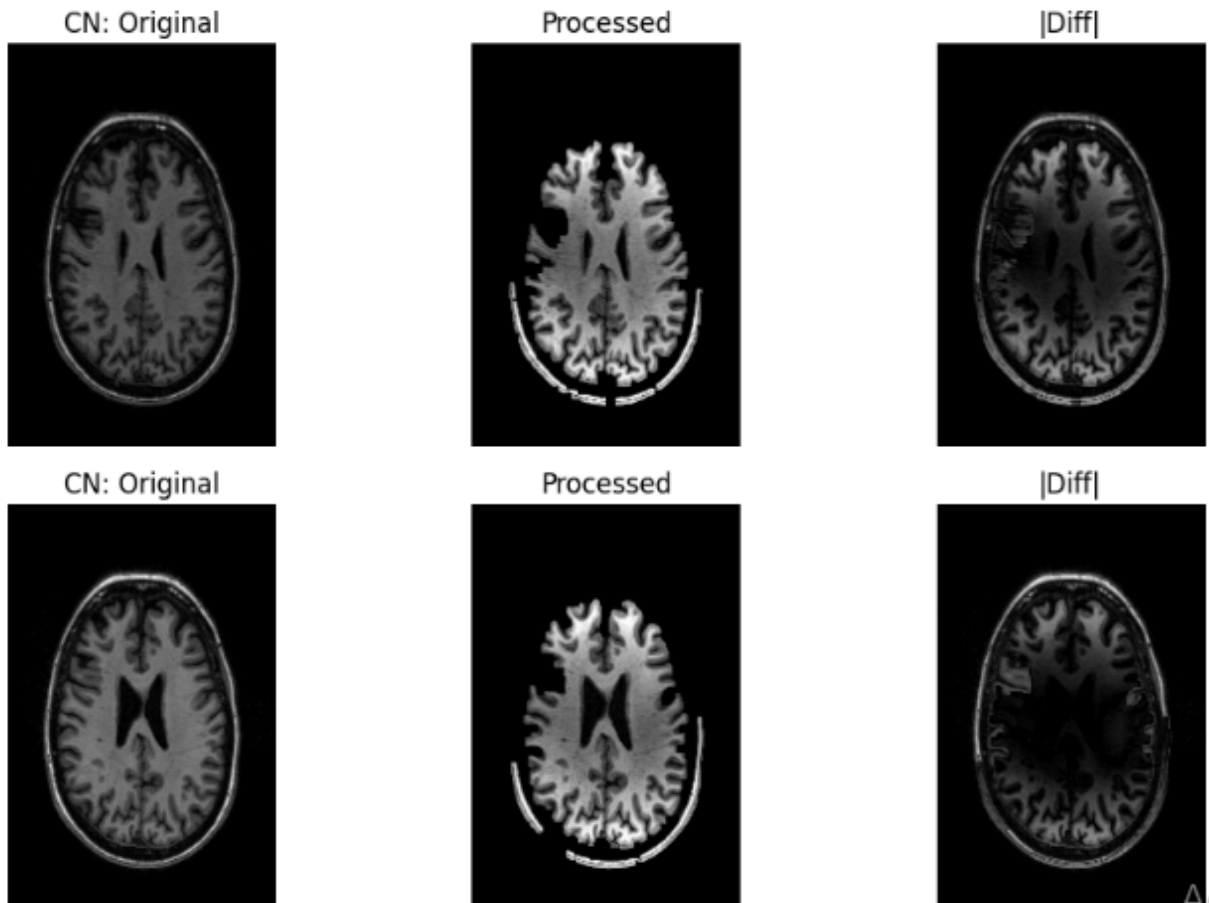


Original: Displays the raw brain scan of a patient diagnosed with Alzheimer's Disease.

Processed: Shows the image after preprocessing steps like normalization, resizing, or filtering.

Difference: Highlights the changes between the original and processed image, often emphasizing regions affected by AD such as hippocampal shrinkage or cortical thinning.

CN Original- Processed- Difference

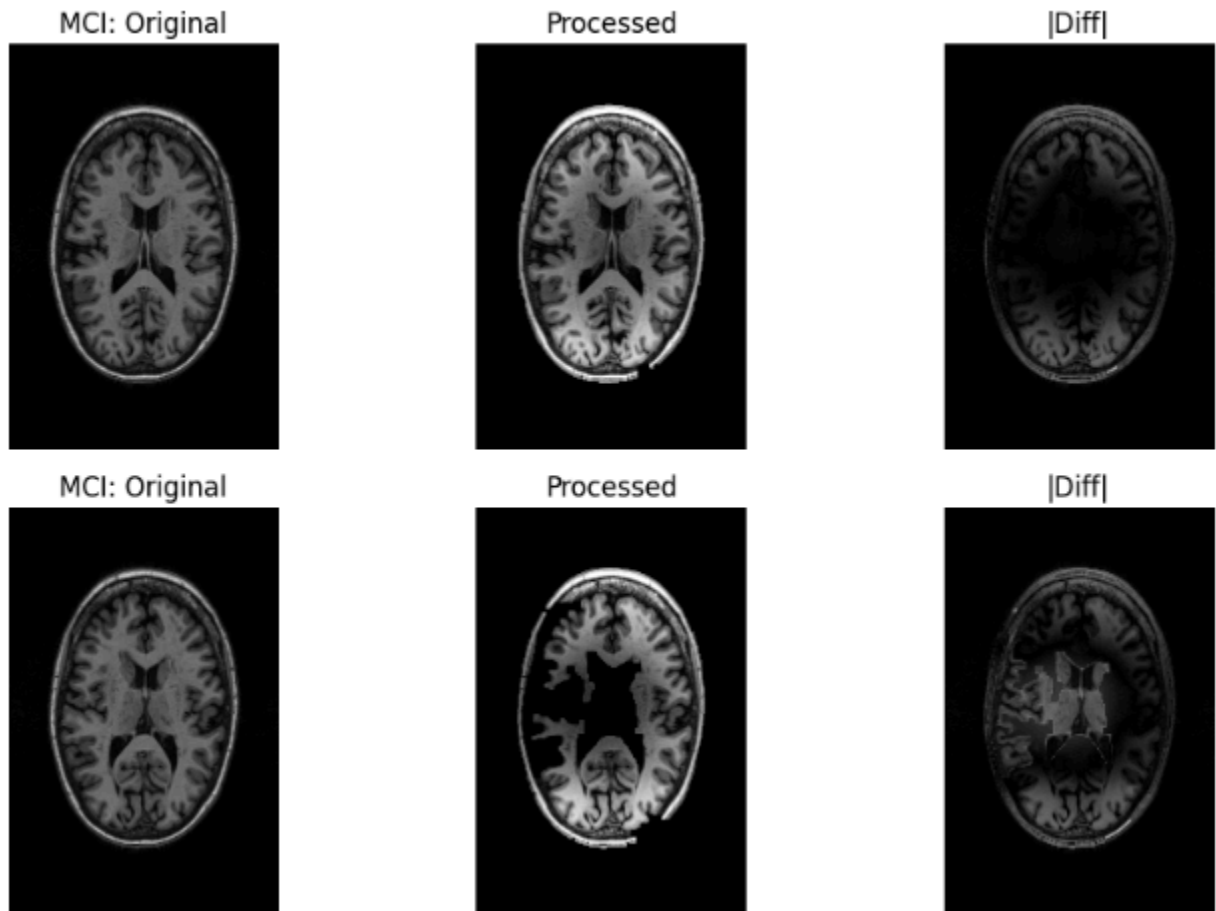


Original: A clean brain scan of a cognitively normal individual.

Processed: The image after enhancement and standardization for model input.

Difference: Minimal changes are visible, indicating stable brain structure with no signs of degeneration.

MCI Original- Processed- Difference



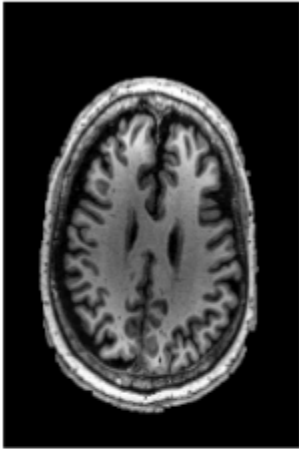
Original: Brain scan of a patient with Mild Cognitive Impairment.

Processed: Preprocessed version optimized for feature extraction.

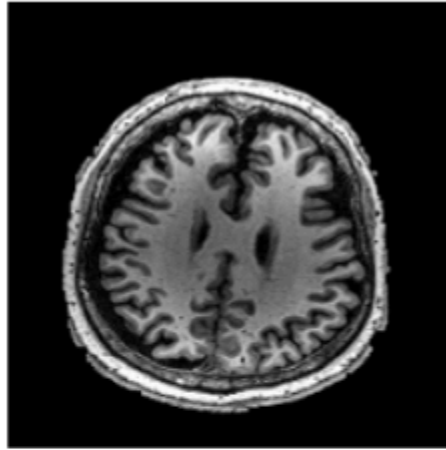
Difference: Shows subtle structural changes, such as mild atrophy, which are crucial for early detection and classification.

Original (256x170) vs. Resized (224x224) images for AD, CN, and MC

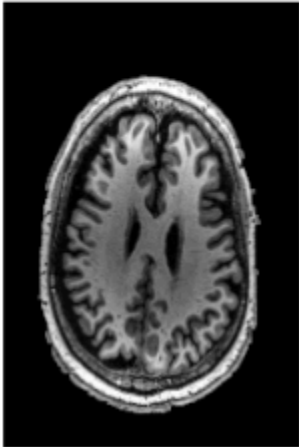
AD: Original (256, 170)



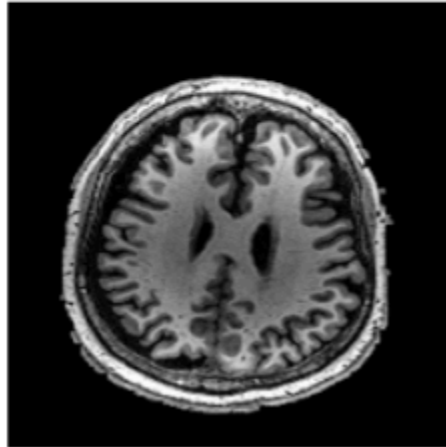
Resized (224, 224)



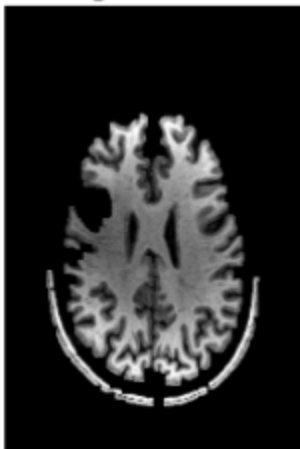
AD: Original (256, 170)



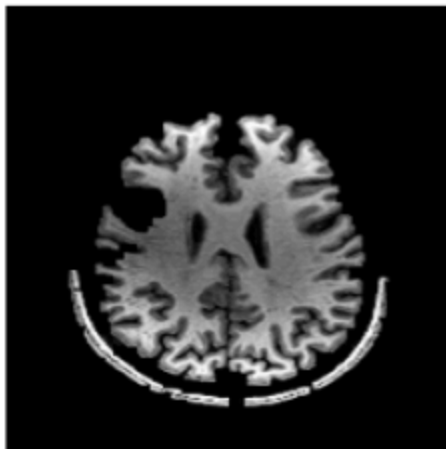
Resized (224, 224)



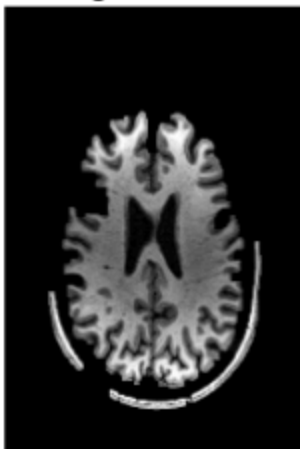
CN: Original (256, 170)



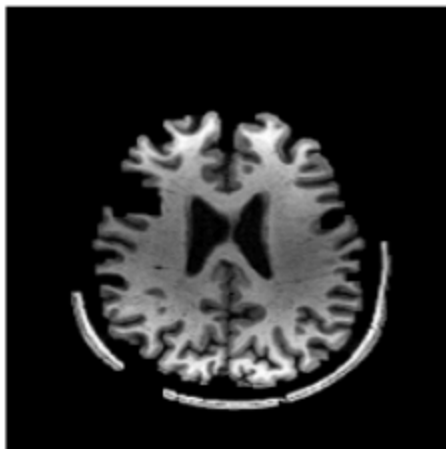
Resized (224, 224)



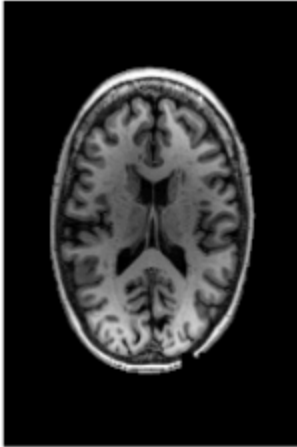
CN: Original (256, 170)



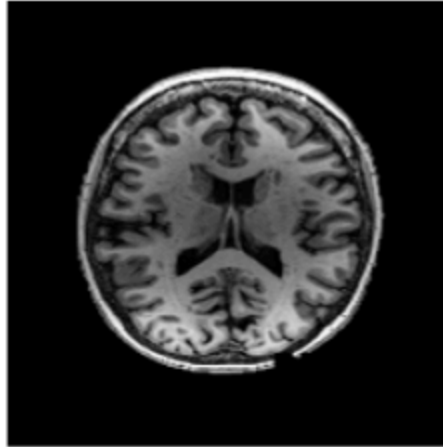
Resized (224, 224)



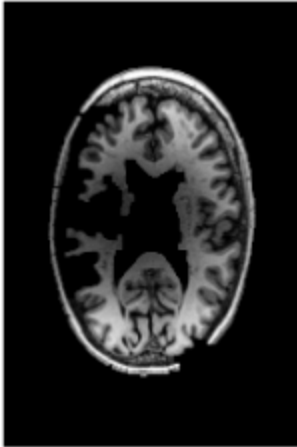
MCI: Original (256, 170)



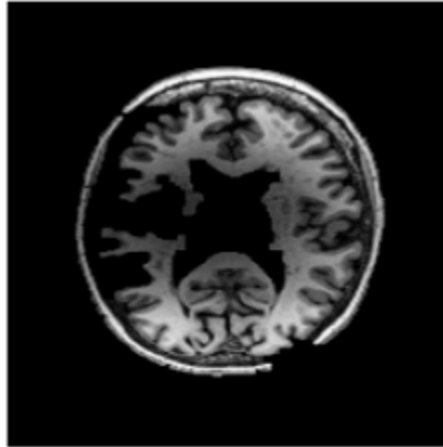
Resized (224, 224)



MCI: Original (256, 170)



Resized (224, 224)



These images demonstrate standardization of input dimensions for deep learning models.

MobileNetV3-large Model

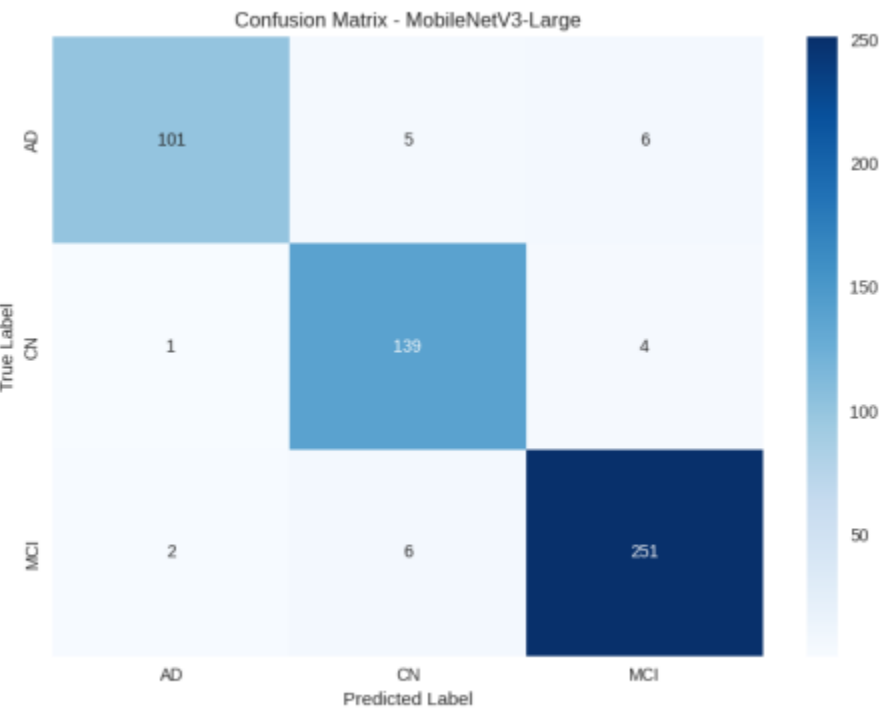
=== MobileNetV3-Large Detailed Results ===

Accuracy: 0.9534
Precision: 0.9540
Recall: 0.9534
F1-Score: 0.9533

Classification Report:

	precision	recall	f1-score	support
AD	0.97	0.90	0.94	112
CN	0.93	0.97	0.95	144
MCI	0.96	0.97	0.97	259
accuracy			0.95	515
macro avg	0.95	0.95	0.95	515
weighted avg	0.95	0.95	0.95	515

Average Prediction Confidence: 0.8284



This graph A 3x3 matrix showing true vs. predicted labels. Most predictions are correct, especially for CN and MCI. AD had 103 correct predictions, with some misclassified as CN and MCI. Indicates good but not perfect classification.



This Bar graph comparing confidence of correct vs. incorrect predictions. Correct predictions cluster around high confidence (0.8–1.0). Incorrect ones are more spread out, often with lower confidence. Shows the model is more confident when it's correct.

EfficientNet V2-S

=== EfficientNetV2-S Detailed Results ===

Accuracy: 0.9786

Precision: 0.9786

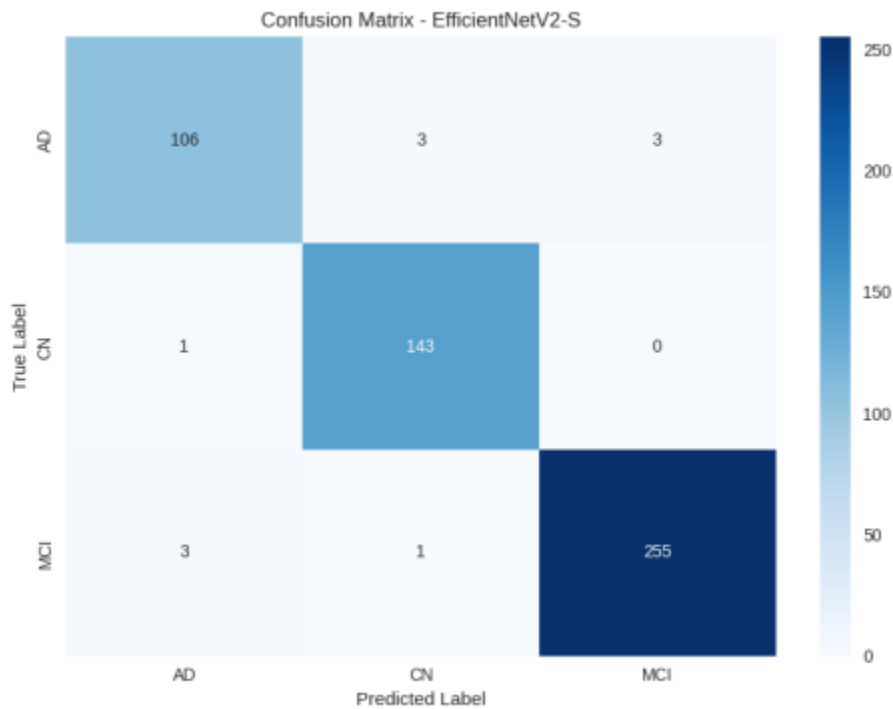
Recall: 0.9786

F1-Score: 0.9786

Classification Report:

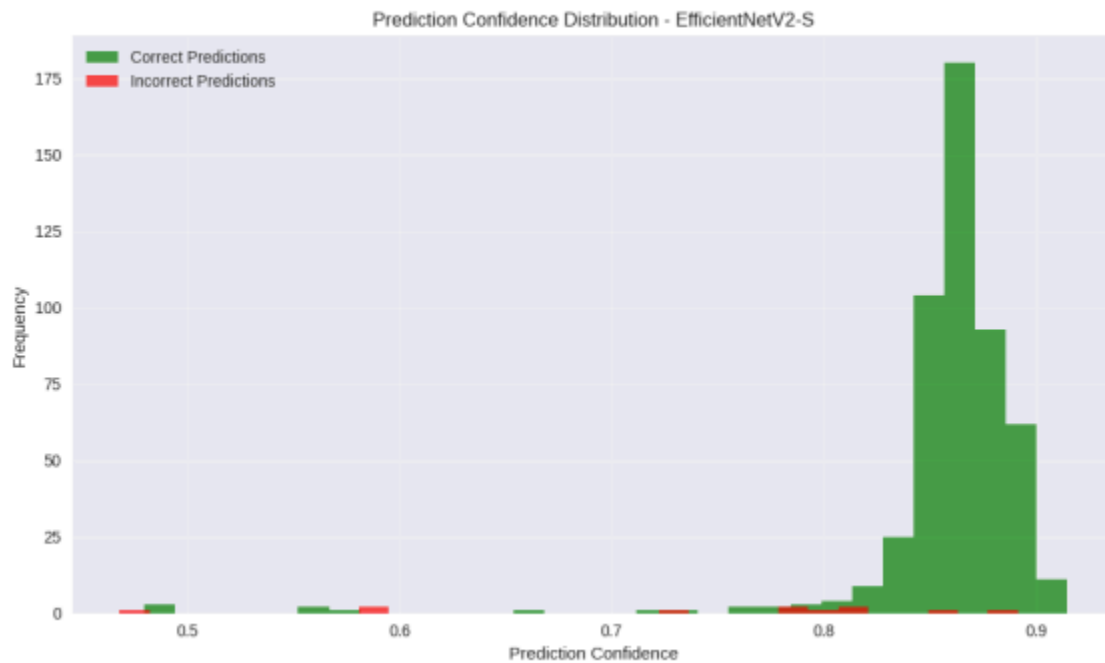
	precision	recall	f1-score	support
AD	0.96	0.95	0.95	112
CN	0.97	0.99	0.98	144
MCI	0.99	0.98	0.99	259
accuracy			0.98	515
macro avg	0.97	0.97	0.97	515
weighted avg	0.98	0.98	0.98	515

Average Prediction Confidence: 0.8568



This graph shows fewer misclassifications than MobileNet. MCI class has 257 correct predictions

out of 259. AD and CN also show improved classification accuracy. Demonstrates stronger performance overall.



This graph shows Higher frequency of confident predictions (0.8 -- 1.0 range).Correct predictions dominate the higher confidence bins.Fewer low-confidence predictions compared to MobileNet.Reflects improved model certainty.

Ensemble Model

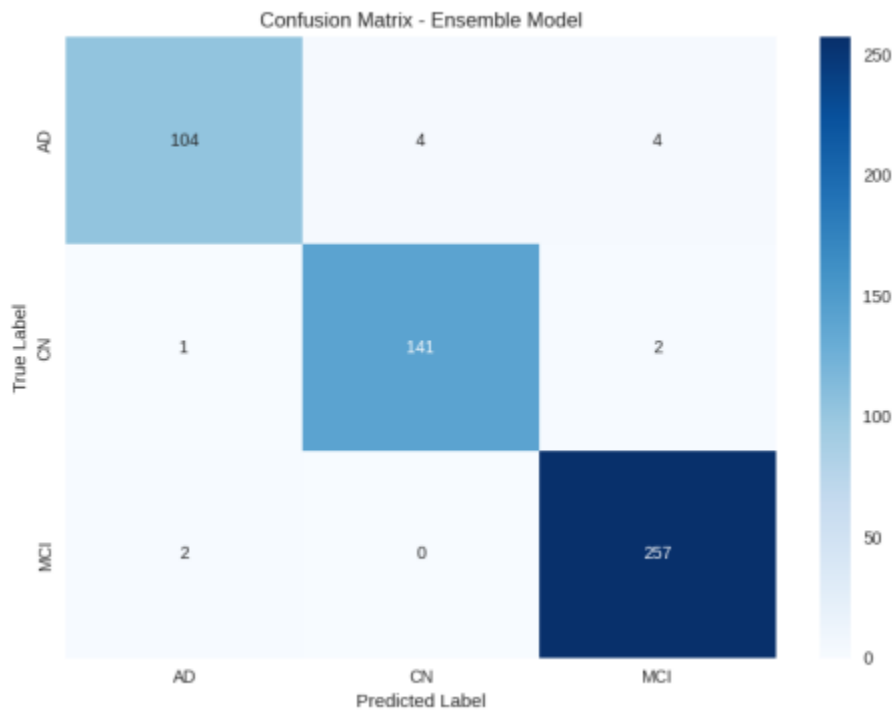
===Ensemble Model Detailed results ===

Accuracy: 0.9748
Precision: 0.9747
Recall: 0.9748
F1-Score: 0.9746

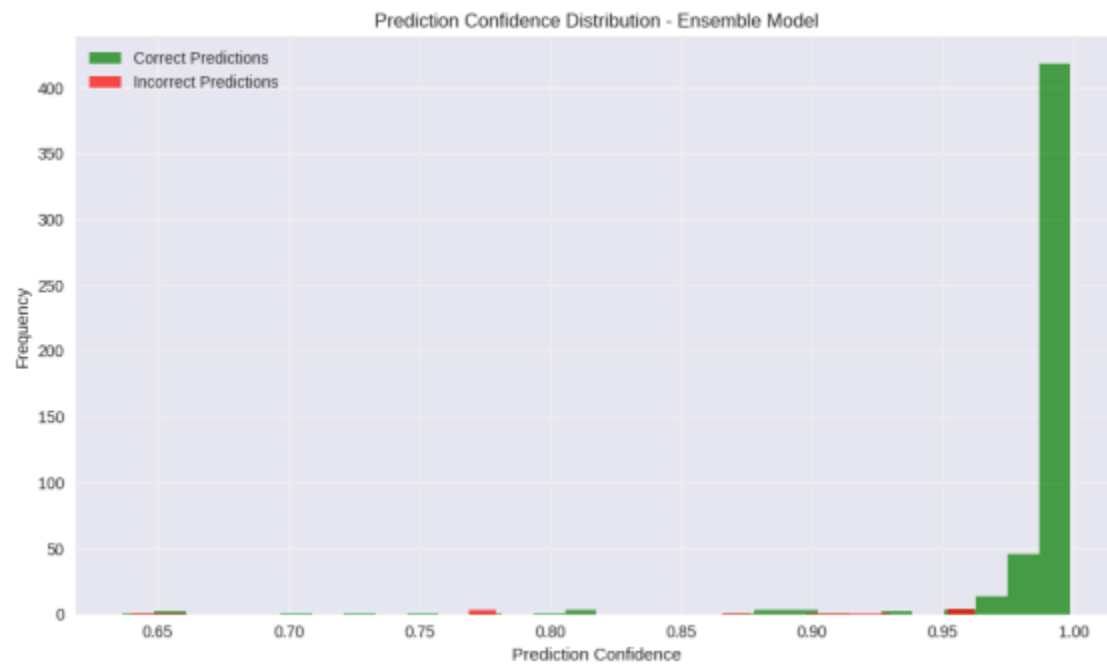
Classification Report:

	precision	recall	f1-score	support
AD	0.97	0.93	0.95	112
CN	0.97	0.98	0.98	144
MCI	0.98	0.99	0.98	259
accuracy			0.97	515
macro avg	0.97	0.97	0.97	515
weighted avg	0.97	0.97	0.97	515

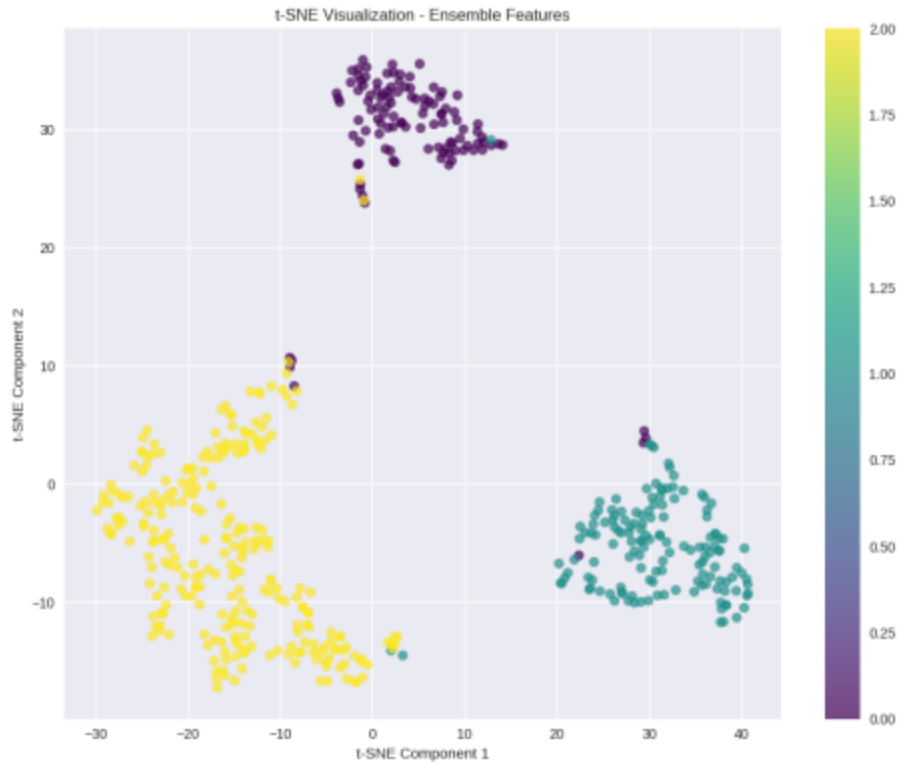
Average Prediction Confidence: 0.9798



This graph shows near-perfect classification across all classes. Only 6 total misclassifications out of 515 samples. MCI class has 100% accuracy. Demonstrates the power of ensemble learning.



This graph given most predictions has confidence above 0.9. Very few incorrect predictions, mostly at lower confidence. Shows the ensemble model is both accurate and confident. Best distribution among all models.



This displays a 2D scatter plot of high-dimensional features. Clear separation between AD, CN, and MCI clusters. Indicates strong feature learning and class distinction. Supports the ensemble model's high performance.

Final Ensemble Results Summary:-

MobileNetV3-Large:

Accuracy: 0.9534

F1-Score: 0.9533

Avg Confidence: 0.8284

EfficientNetV2-S:

Accuracy: 0.9786

F1-Score: 0.9786

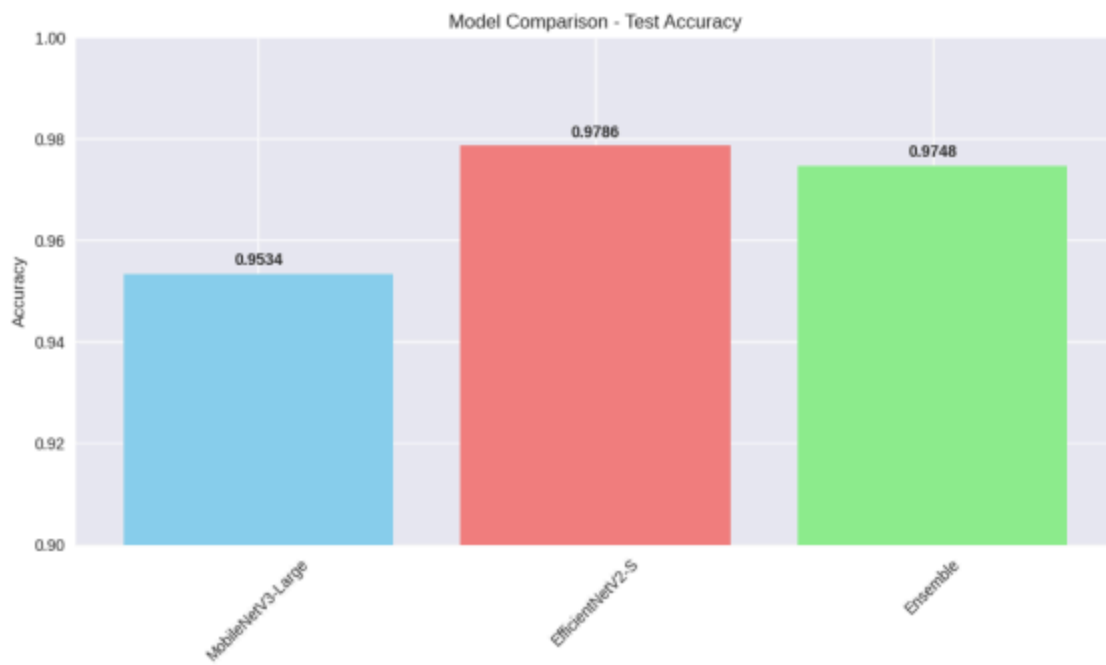
Avg Confidence: 0.8568

Ensemble:

Accuracy: 0.9748

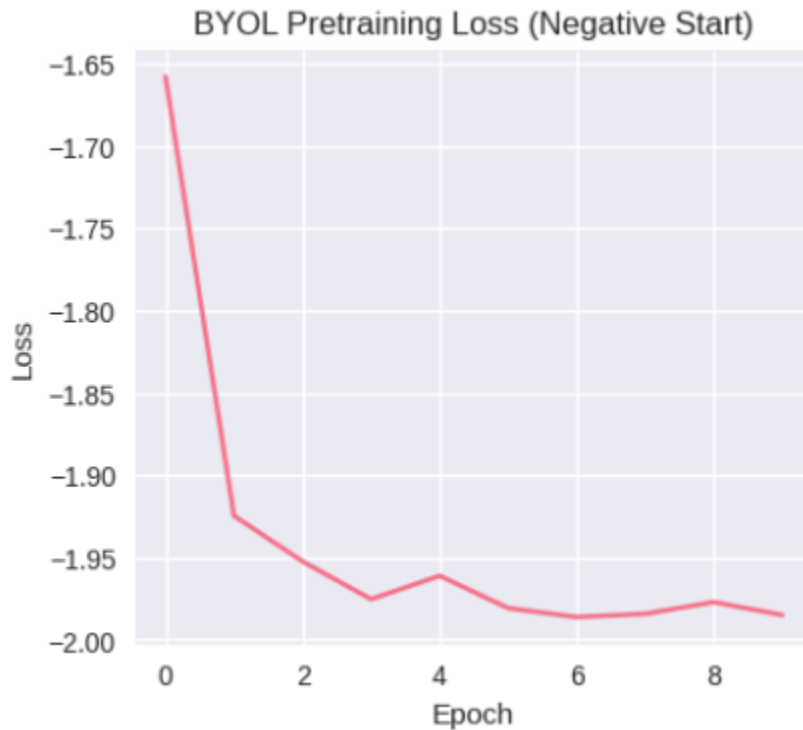
F1-Score: 0.9746

Avg Confidence: 0.9798

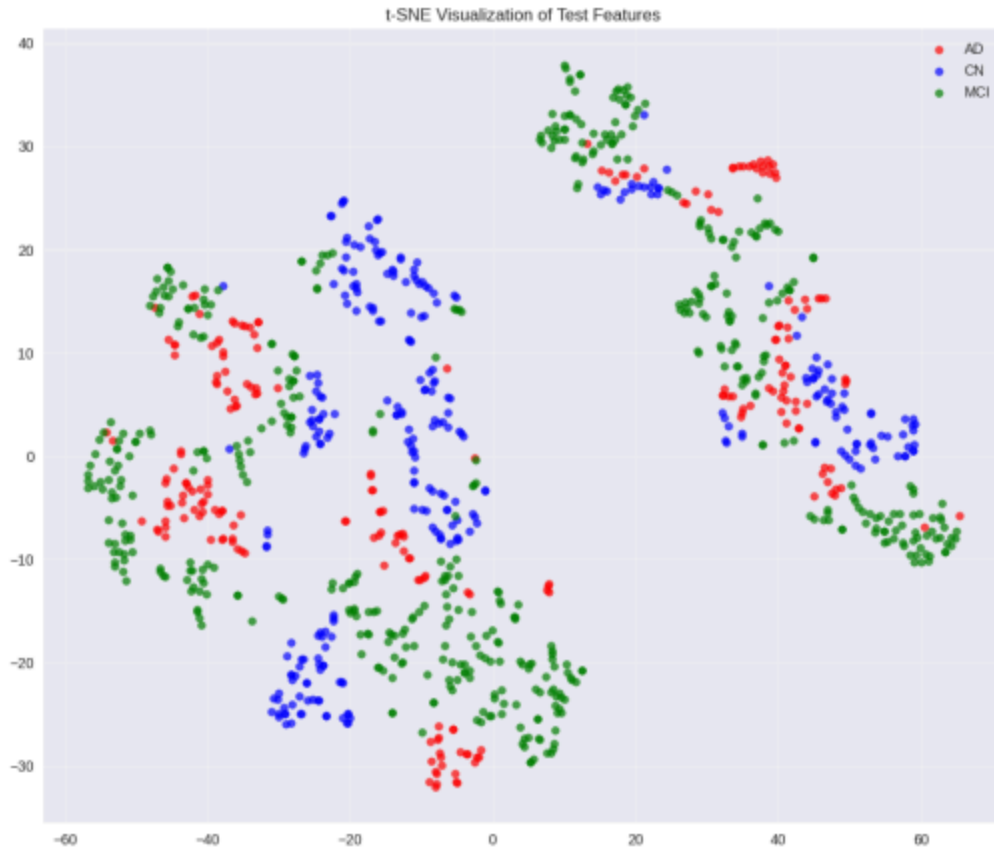


This Bar chart compares MobileNet, EfficientNet, and Ensemble. Ensemble outperforms both in accuracy and F1-score. Highlights the benefit of combining models. Visual summary of performance metrics.

SSL-1 BYOL Method



This graph shows a **BYOL pretraining loss curve** labeled “Negative Start.” The y-axis represents **loss values** ranging from -1.65 to -2.00. The x-axis represents **epochs** from 0 to 8. The curve indicates a **decreasing trend in loss** over epochs. Suggests successful optimization during pretraining. Below the graph, there is a **t-SNE visualization** of test features. Data points are scattered across a 2D plane with axes from -60 to 60. Points are color-coded for categories: **AD**, **CN**, **MCI**. Indicates clustering patterns among different classes. Used for evaluating feature separability after training.



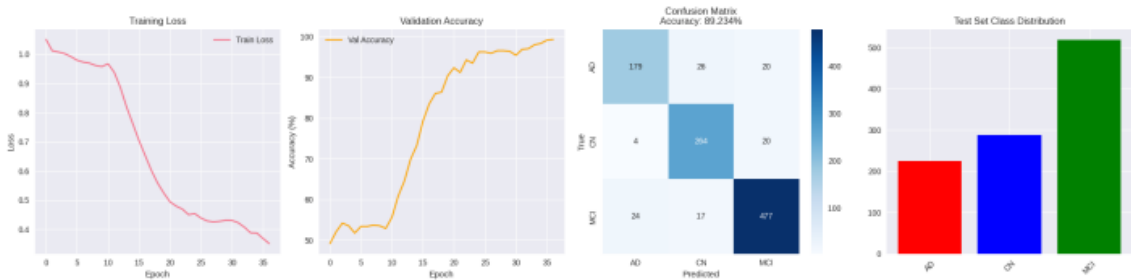
This displays **training loss and validation accuracy curves** over epochs. The x-axis shows epochs (0–30), y-axis shows respective metrics. Training loss decreases steadily, while validation accuracy improves. Includes a **confusion matrix** for classification performance. Accuracy reported as approximately **89.23%**. Shows **test set class distribution** with bar plots. Bars represent counts for different classes (values like 49, 47). Indicates balanced or slightly imbalanced dataset. Confusion matrix suggests good prediction alignment with true labels. Overall, the image summarizes model performance and evaluation.

Final Evaluation

TTA Evaluation: 100% | 33/33 [00:12<00:00, 2.62it/s]

ENHANCED FINAL ACCURACY: 89.234%

	precision	recall	f1-score	support
AD	0.86	0.80	0.83	225
CN	0.86	0.92	0.89	288
MCI	0.92	0.92	0.92	518
accuracy			0.89	1031
macro avg	0.88	0.88	0.88	1031
weighted avg	0.89	0.89	0.89	1031



Detailed Metrics:

Weighted Precision: 0.8925

Weighted Recall: 0.8923

Weighted F1-Score: 0.8918

Per-class Metrics:

AD: Precision=0.8647, Recall=0.7956, F1=0.8287

CN: Precision=0.8599, Recall=0.9167, F1=0.8874

MCI: Precision=0.9226, Recall=0.9208, F1=0.9217